

GATE CS 2027: COMPLETE PREPARATION BLUEPRINT

Organizing Institute: IIT Madras | 21 Sep 2026 – Feb 2027 | 20 Weeks (139 Days) | AIR < 100 System

1. THE 6-HOUR/DAY HIGH-YIELD ENGINE (834 Hours Total)

Why 6 Hours is the Optimal Sweet Spot: 139 Days × 6 Hours = 834 high-intensity hours. Divided into four structured blocks, it avoids burnout while allowing 380h concept mastery, 240h PYQs (3,200+ questions), 120h for 2 full revisions, and 94h for 10 full-length mocks & 4-hour post-mortems.

Time Slot	Session Focus	Activity & Deliverable	Locked Resource
07:00 – 09:00 AM (2.0h)	Track A: Concept Study	Watch theory lectures + write 1-page condensed notes & derivations.	Neso Academy / GO Classes
09:30 – 10:30 AM (1.0h)	Track A: PYQ Grind	Solve 15 GateOverflow PYQs independently on paper (No peeking).	GateOverflow Question Bank
03:00 – 05:00 PM (2.0h)	Track B: Concept Study	Watch systems lecture (Logic/TOC/COA/OS/DBMS/CN) + handwritten traces.	Vishwadeep Gothi / Amit Khurana
06:00 – 07:00 PM (1.0h)	Track B: PYQs & Logging	Solve 10–12 PYQs + log wrong answers into Mistake Notebook + 15m Aptitude.	GateOverflow + Error Vault

2. 4-PHASE TIMELINE & STRATEGIC MILESTONES

Phase	Dates & Duration	Core Target	Target Deliverables
Phase 1: 1st Syllabus Pass	21 Sep – 27 Dec 2026 (14 Weeks)	100% Official IIT Madras Syllabus coverage.	All 10 subjects + 35-yr PYQs + Short Notes
Phase 2: Revision 1 & Tests	28 Dec 2026 – 10 Jan 2027 (2 Weeks)	Consolidation + Re-solve Red PYQs.	Complete 1st Revision + Multi-subject Tests
Phase 3: Mocks & Revision 2	11 Jan – 31 Jan 2027 (3 Weeks)	Full Exam simulation + Post-mortems.	10 Full-length CBT Mocks + 2nd Revision
Phase 4: Final Climax	01 Feb – Exam Day 2027 (1 Week)	Peak mental clarity & zero new topics.	Formula sheets + Mistake review + Calibrations

3. COMPLETE 20-WEEK CALENDAR (WEEKS 1 TO 20)

Week & Dates	Track A / Major Focus	Track B / Co-Subject	Target	Milestone / Test
Week 1 (21–27 Sep)	C Programming (Operators, Control, Pointers, Recursion)	Discrete Math: Propositional & First Order Logic, Quantifiers	60 PYQs	Topic Test 1 (C + Logic)
Week 2 (28 Sep–04 Oct)	Linear DS (1D/2D Arrays Row/Col Major, Stacks, Queues, LL)	Discrete Math: Sets, Relations, Equivalence, POSETs, Lattices	70 PYQs	Milestone Test 1 (C + DS + Sets)
Week 3 (05–11 Oct)	Non-Linear DS (Binary Trees, BST, Heaps, Heapify, Heap Sort)	Discrete Math: Monoids, Groups, Lagrange, Combinatorics	80 PYQs	Topic Test (Trees, Heaps, Groups)
Week 4 (12–18 Oct)	Algorithms (Asymptotics $O/\Omega/\Theta$, Master Theorem, Sorting)	Discrete Math: Recurrences, Generating Func, Graph Theory	85 PYQs	Discrete Math & DS Complete Test
Week 5 (19–25 Oct)	Algorithms (Divide & Conquer, Greedy: Huffman, Knapsack, MST)	Linear Algebra: Systems $AX=B$, Determinants, Rank, LU	80 PYQs	Greedy & Linear Algebra Test
Week 6 (26 Oct–01 Nov)	Algorithms (Dynamic Prog: LCS, MCM, Knapsack, Graphs, Hashing)	Linear Algebra (Eigenvalues, Cayley) + Digital (K-Maps, Tabular)	90 PYQs	Algorithms & Linear Algebra Seal
Week 7 (02–08 Nov)	Digital Logic (Adders, MUX, Decoders, Flip-Flops, Counters, IEEE 754)	TOC: DFA Design, Minimization, NFA, Regular Expressions	85 PYQs	Digital Logic Full Subject Test
Week 8 (09–15 Nov)	TOC (Pumping Lemma, Closure, Grammars, Pushdown Automata, CFLs)	Calculus: Limits, Continuity, Differentiability, Maxima, Integrals	80 PYQs	TOC & Calculus Topic Test
Week 9 (16–22 Nov)	TOC (Turing Machines, Chomsky Hierarchy, Undecidability, PCP, Rice)	Compiler Design: Lexical Analysis, LL(1), LR(0), SLR, LALR, CLR	85 PYQs	TOC Full Subject Test
Week 10 (23–29 Nov)	Compiler Design (SDT, 3-Address Code, DAG, Optimization, Data Flow)	Probability & Stats: Bayes, Random Vars, Distributions, StdDev	90 PYQs	Compiler & Math Grand Test
Week 11 (30 Nov–06 Dec)	COA (Instruction Formats, Addressing Modes, ALU, Control Unit)	OS: Process States, PCB, Threads, System Calls, CPU Scheduling	90 PYQs	ISA & Scheduling Test
Week 12 (07–13 Dec)	COA (Cache Direct/Assoc/Set-Assoc, EMAT, I/O DMA, Pipelining)	OS: Concurrency, Semaphores, Deadlocks, Paging, Inode Files	100 PYQs	COA & OS Full Subject Tests
Week 13 (14–20 Dec)	DBMS (ER Model, Schema, Relational Algebra, SQL, Normalization)	Computer Networks: Switching, Delays, Framing, CRC, MAC, CSMA/CD	95 PYQs	DBMS & CN Data Link Test

Week 14 (21–27 Dec)	DBMS (B/B+ Trees, Transactions, ACID, Serializability, 2PL)	Computer Networks: IPv4, CIDR, Routing, TCP, Congestion, DNS/HTTP	100 PYQs	100% 1ST PASS COMPLETE TEST
Week 15 (28 Dec–03 Jan)	Phase 2: Revision 1 (Math + DS + Algo + Digital + COA)	Re-solve all Red/Yellow PYQs + Short notes review	100 Re-PYQs	Combined Multi-Subject Test 1
Week 16 (04–10 Jan)	Phase 2: Revision 1 (TOC + Compiler + OS + DBMS + CN + Aptitude)	Re-solve all Red/Yellow PYQs + Mistake vault review	100 Re-PYQs	Combined Multi-Subject Test 2
Week 17 (11–17 Jan)	Phase 3: CBT Full Mock 1 (GATE 2023), Mock 2 & Mock 3 (09:30 AM)	4-Hour Post-Mortem error audit per mock + 2nd Revision pass	3 Full Mocks	Mock Post-Mortem 1, 2, 3
Week 18 (18–24 Jan)	Phase 3: CBT Full Mock 4 (GATE 2024), Mock 5 & Mock 6 (09:30 AM)	4-Hour Post-Mortem + Speed calibration + NAT bounds check	3 Full Mocks	Mock Post-Mortem 4, 5, 6
Week 19 (25–31 Jan)	Phase 3: CBT Full Mock 7 (GATE 2025), Mock 8 & Mock 9 (GATE 2026)	Final 2nd Revision seal + Master formula sheet audit	3 Full Mocks	Final Benchmark Mock 9
Week 20 (01 Feb–Exam)	Phase 4: Final 7-Day Protocol — Zero new study, formula runs	Mistake notebook review + Virtual calc drills + Sleep sync	Formula Drills	GATE 2027 EXAM DAY (AIR < 100)

4. LOCKED RESOURCES DIRECTORY (ALL 10 SECTIONS)

Subject	Primary Resource (Free)	Backup Resource	PYQ & Practice Source
Discrete Math	GO Classes by Deepak Poonia & Sachin Mittal	Amit Khurana Sir	GateOverflow Discrete Math
Linear Algebra	GO Classes / Gilbert Strang MIT 18.06	Gajendra Purohit	GateOverflow Linear Algebra
Calculus & Prob	GO Classes Probability & Calculus	Unacademy CS	GateOverflow Engg Math
Digital Logic	Neso Academy Digital Electronics	Amit Khurana Sir	GateOverflow Digital Logic
C & Data Struct.	Neso Academy (C) + Abdul Bari (DS)	Amit Khurana Sir	GateOverflow Programming & DS
Algorithms	Abdul Bari Algorithms Playlist	GO Classes Algorithms	GateOverflow Algorithms
Theory of Comp.	GO Classes TOC by Deepak Poonia	Amit Khurana Sir	GateOverflow TOC
Compiler Design	Gate Smashers / GO Classes Compiler	Amit Khurana Sir	GateOverflow Compiler Design
COA	Vishwadeep Gothi Sir (Unacademy CS)	GO Classes COA	GateOverflow COA
Operating Systems	Vishwadeep Gothi Sir / Amit Khurana	Gate Smashers OS	GateOverflow Operating Systems
DBMS	Amit Khurana Sir / GO Classes DBMS	Vishwadeep Gothi Sir	GateOverflow Databases
Comp. Networks	Ankit Doyla Sir / Amit Khurana	Gate Smashers CN	GateOverflow Computer Networks
General Aptitude	Saurabh Thakur Sir / GATE Wallah	GFG Aptitude	GateOverflow Aptitude (2010–2026)

5. MISTAKE NOTEBOOK & EXAM EXECUTION PROTOCOL

The 4 Error Categories: [TYPE-A: Concept Gap] Re-watch lecture segment and add concept definition to notes. [TYPE-B: Trap / Misinterpretation] Write exact trap wording in red bold. [TYPE-C: Calculation / NAT Slip] Repeat calculation 3 times on rough paper. [TYPE-D: Formula Amnesia] Add directly to 1-page formula sheet.

3-Round Exam Execution: Round 1 (0–75m: Direct 1-mark & easy questions, secure 35+ marks) → Round 2 (75–145m: Standard 2-mark numericals, add +30 marks) → Round 3 (145–180m: Tough MSQs & precision NAT checks).